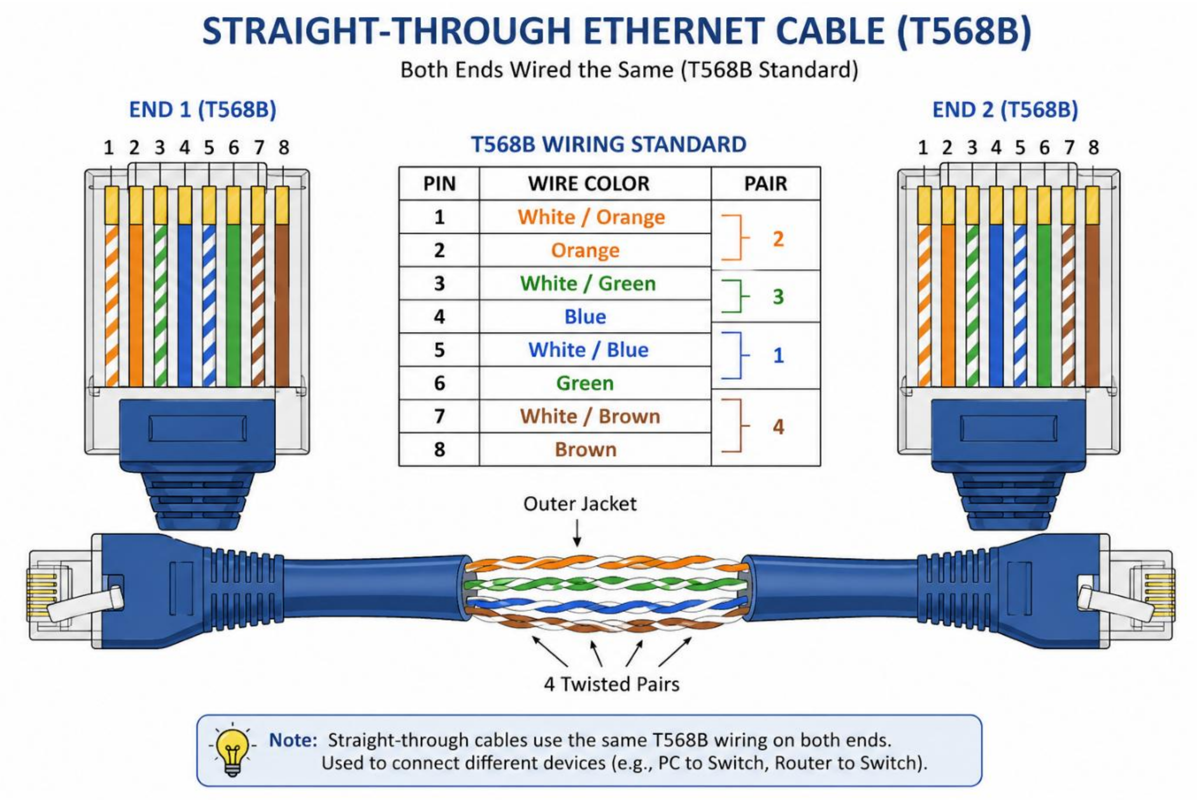
1) List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

Feature	Ethernet Cable	Fiber Optic Cable
1. Transmission medium	Uses copper wires to transmit electrical signals.	Uses glass/plastic fibers to transmit light signals.
2. Speed & distance	Good speed but generally suited to shorter distances (commonly up to 100 m for standard Ethernet runs).	Supports very high speeds and much longer distances.
3. Interference	Can be affected by electromagnetic interference (EMI).	Resistant to electromagnetic interference, making it suitable for electrically noisy environments.

2) **Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.**



3) Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

Watch The Youtube Video :- https://www.youtube.com/watch?v=NWhoJp8UQpo

Steps:

1. **Cut the CAT6 cable**
   Cut the cable to the required length using a cable cutter. Make sure the end is clean and straight.

2. **Remove the outer jacket**
   Use a cable stripper to carefully remove about 2–3 cm of the outer insulation. Do not damage the twisted wires inside.

3. **Separate the four twisted pairs**
   Untwist the orange, green, blue, and brown pairs. Straighten the individual wires gently.

4. **Arrange the wires in T568B order**
   From **pin 1 to pin 8**, arrange the wires as follows:

Pin	Wire color
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

5. **Straighten and trim the wires**
   Hold all eight wires together in the correct order and straighten them. Trim the ends evenly so they can enter the RJ45 connector properly.

6. **Insert the wires into the RJ45 connector**
   Hold the RJ45 plug with the **gold contacts facing upward** and the cable opening toward you. Carefully push all eight wires into the connector, maintaining the T568B order. Make sure each wire reaches the front of the connector.

7. **Check the wiring**
   Before crimping, verify that the wires are still in the correct order from **pin 1 through pin 8** and that the cable jacket enters the connector far enough to be secured.

8. **Crimp the connector**
   Place the RJ45 plug into the appropriate socket of the crimping tool. Squeeze the handles firmly so the metal contacts are pressed into the eight wires and the cable is held securely.

9. **Inspect the finished connector**
   Check that all eight contacts are properly seated and that the cable jacket is held by the connector's strain relief.

10. **Test the cable**
    Connect both ends of the completed cable to an Ethernet cable tester. The tester should show pins **1–8 in the correct sequence**. For a straight-through cable, use **T568B on both ends**.

Result:
The CAT6 cable is now terminated with an RJ45 connector and can be used for Ethernet connections such as connecting a **computer to a switch or router**.

4) **Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

Hint: Consider speed, distance, and budget for your answer.**

For a gaming café with 10 PCs, I would use Ethernet (CAT6) cables because they provide sufficient speed for LAN gaming and are cost-effective for short distances. I would use an RJ45 crimper to attach RJ45 connectors to the CAT6 cables. Fiber optic cables and a fiber cleaver would be more expensive and are generally unnecessary for connecting PCs within a small café. Ethernet is therefore the better choice because it offers a good balance of speed, distance, ease of installation, and budget.